

Random Forest Classifier on MNIST - Experiment Report

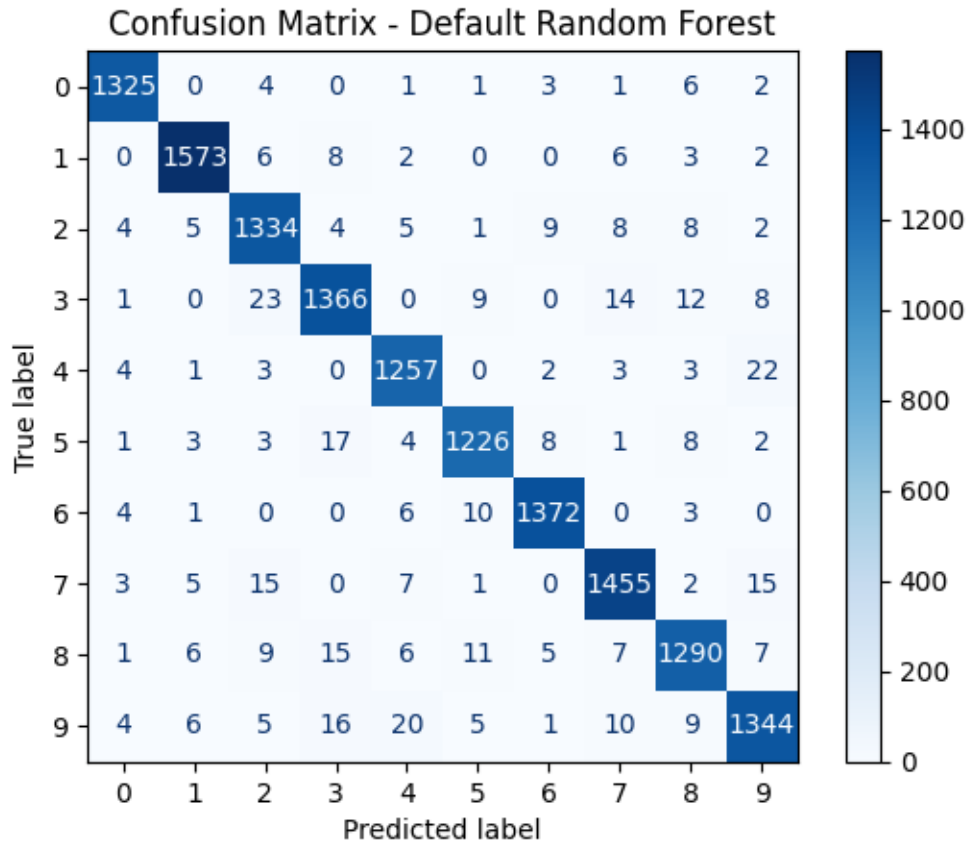
This report summarizes the experiments conducted using a Random Forest Classifier on the MNIST handwritten digit dataset. Three different hyperparameter settings were tested, and their performance was evaluated based on training time, accuracy, and confusion matrices.

Experiment 1: Default Hyperparameters

- **Hyperparameters:** *n_estimators=100, max_depth=None, min_samples_split=2, min_samples_leaf=1, max_features='sqrt'* (default values)
- **Training Time:** 58.24 seconds
- **Training Accuracy:** 1.0
- **Testing Accuracy:** 0.967
- **Classification Report (Test Set):**

Classification Report (default):				
	precision	recall	f1-score	support
0	0.98	0.99	0.99	1343
1	0.98	0.98	0.98	1600
2	0.95	0.97	0.96	1380
3	0.96	0.95	0.96	1433
4	0.96	0.97	0.97	1295
5	0.97	0.96	0.97	1273
6	0.98	0.98	0.98	1396
7	0.97	0.97	0.97	1503
8	0.96	0.95	0.96	1357
9	0.96	0.95	0.95	1420
accuracy			0.97	14000
macro avg	0.97	0.97	0.97	14000
weighted avg	0.97	0.97	0.97	14000

- **Confusion Matrix (Test Set):**



The default model achieved perfect accuracy on the training set but showed a slight drop in performance on the test set, indicating some overfitting. The confusion matrix shows that the model performed well across most digits, with a few misclassifications occurring between visually similar digits like 3 and 8, and 4 and 9.

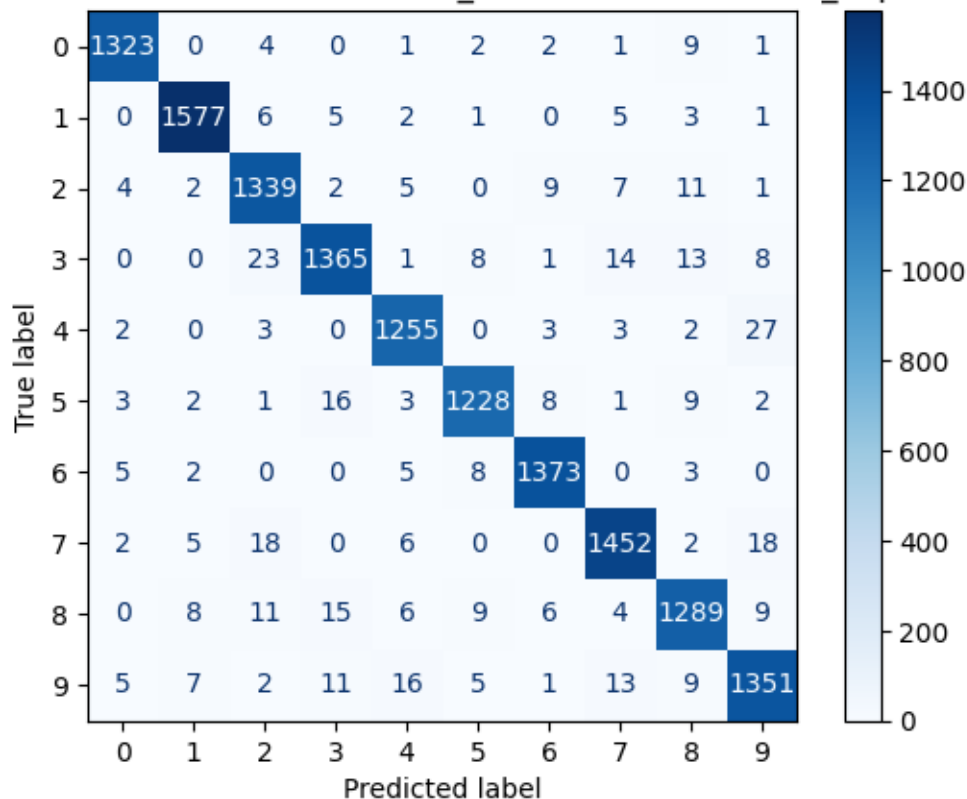
Experiment 2: Custom Hyperparameters ($n_estimators=200$, $max_depth=20$)

- **Hyperparameters:** $n_estimators=200$, $max_depth=20$, $min_samples_split=2$, $min_samples_leaf=1$, $max_features='sqrt'$
- **Training Time:** 103.68 seconds
- **Training Accuracy:** 0.999
- **Testing Accuracy:** 0.968
- **Classification Report (Test Set):**

Classification Report (custom 1):					
	precision	recall	f1-score	support	
0	0.98	0.99	0.98	1343	
1	0.98	0.99	0.98	1600	
2	0.95	0.97	0.96	1380	
3	0.97	0.95	0.96	1433	
4	0.97	0.97	0.97	1295	
5	0.97	0.96	0.97	1273	
6	0.98	0.98	0.98	1396	
7	0.97	0.97	0.97	1503	
8	0.95	0.95	0.95	1357	
9	0.95	0.95	0.95	1420	
accuracy			0.97	14000	
macro avg	0.97	0.97	0.97	14000	
weighted avg	0.97	0.97	0.97	14000	

- **Confusion Matrix (Test Set):** (See plot below)

Confusion Matrix - Random Forest (n_estimators=200, max_depth=20)



Increasing the number of estimators and setting a specific max_depth slightly improved the testing accuracy compared to the default model. The training accuracy was still very high,

indicating some overfitting persists. The confusion matrix is similar to the default model, with slight variations in the number of misclassifications for certain digits.

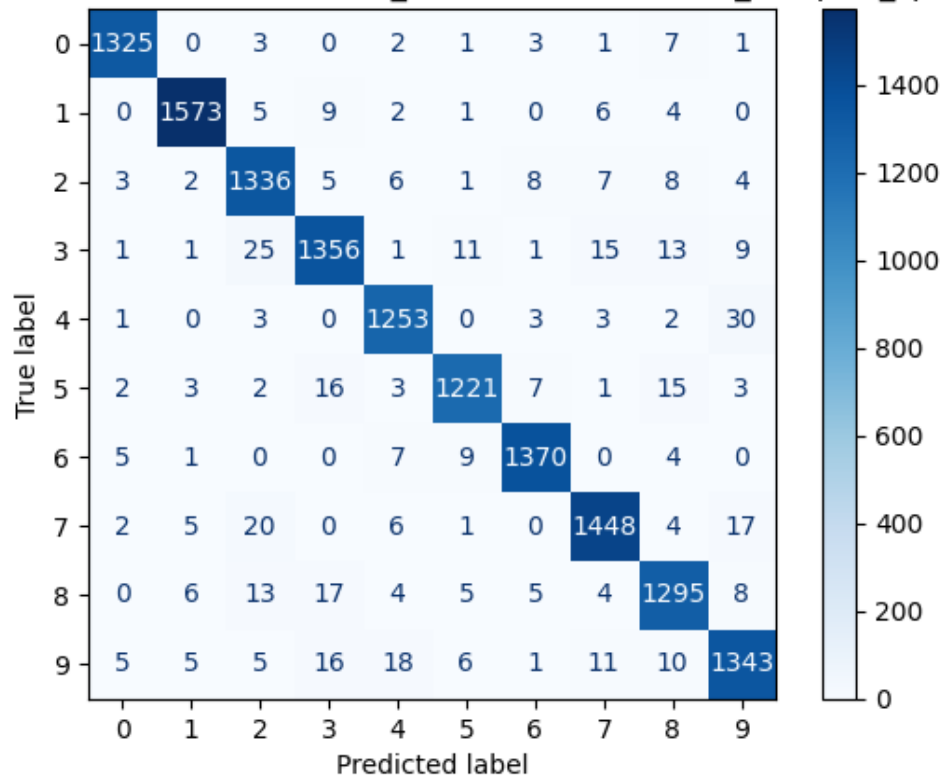
Experiment 3: Custom Hyperparameters (`n_estimators=300`, `min_samples_split=10`)

- **Hyperparameters:** `n_estimators=300`, `max_depth=None`, `min_samples_split=10`, `min_samples_leaf=1`, `max_features='sqrt'`
- **Training Time:** 163.52 seconds
- **Training Accuracy:** 0.9985
- **Testing Accuracy:** 0.9657
- **Classification Report (Test Set):**

Classification Report (custom 2):					
	precision	recall	f1-score	support	
0	0.99	0.99	0.99	1343	
1	0.99	0.98	0.98	1600	
2	0.95	0.97	0.96	1380	
3	0.96	0.95	0.95	1433	
4	0.96	0.97	0.96	1295	
5	0.97	0.96	0.97	1273	
6	0.98	0.98	0.98	1396	
7	0.97	0.96	0.97	1503	
8	0.95	0.95	0.95	1357	
9	0.95	0.95	0.95	1420	
accuracy			0.97	14000	
macro avg	0.97	0.97	0.97	14000	
weighted avg	0.97	0.97	0.97	14000	

- **Confusion Matrix (Test Set):** (See plot below)

Confusion Matrix - Random Forest (n_estimators=300, min_samples_split=10)



This experiment used even more estimators and increased min_samples_split. The training time increased significantly with the higher number of estimators. The testing accuracy was slightly lower than the other two experiments. The training accuracy was still very high, indicating overfitting. The confusion matrix patterns are similar to the other models.

Discussion and Findings

All three Random Forest models performed well on the MNIST dataset, achieving testing accuracies around 96-97%. The models consistently showed higher accuracy on the training set than on the test set, indicating some degree of overfitting, which is common with tree-based models when not heavily regularized.

Increasing the number of estimators (n_estimators) generally led to longer training times but did not result in a substantial improvement in testing accuracy in these experiments.

Setting max_depth to 20 in Experiment 2 resulted in a slightly higher testing accuracy compared to the default and Experiment 3, suggesting that limiting the depth can sometimes help generalize better, although the training accuracy remained very high.

Increasing min_samples_split in Experiment 3 aimed to prevent the trees from splitting on very small groups of samples, which can help reduce overfitting. While the training accuracy slightly decreased, the testing accuracy also saw a minor decrease compared to Experiment 2.

The confusion matrices across all experiments showed similar patterns of misclassification, primarily between digits that share visual similarities. This suggests that the challenges in classification are inherent to the visual nature of the data and the ability of the model to distinguish between subtle variations in handwritten digits.

Overall, while increasing the number of estimators and adjusting `max_depth` and `min_samples_split` did have some impact on training time and slightly on testing accuracy, all three models demonstrated robust performance on the MNIST dataset. Further hyperparameter tuning, potentially including parameters like `min_samples_leaf` and `max_features`, and techniques like cross-validation, could be explored to potentially further optimize performance and reduce overfitting.

Submitted By:

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